Botany Lab

COURSE CODE: 2129 **REVISION: 2021** **SEMESTER: 2**

Diploma in Wood and Paper Technology

Category
Engineering Science

Periods/Week
3 (L:0, T:0, P:3)

Total Periods
45 per Semester

Credits
No Credit

Course Outcomes (COs)

CO No.	Description	Duration	Cognitive Level
CO1	Interpret Plant diversity and morphology of root, stem and leaves	9 Hours	Applying
CO2	Analyse reproductive morphology and morphology of fruits	9 Hours	Applying
CO3	Identify angiosperms Taxonomy	12 Hours	Applying
CO4	Interpret Plant anatomy (Wood Structure focus)	12 Hours	Applying

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Calculations & Measurements

In Botany Lab, calculations are primarily used for microscopic magnification and determining the age of wood samples.

1. Microscopic Magnification

Formula: Total Magnification = Power of Objective Lens × Power of Ocular (Eyepiece) Lens

Example: If the eyepiece is 10x and the objective lens is 40x:

Calculation: $10 \times 40 = 400$x total magnification.

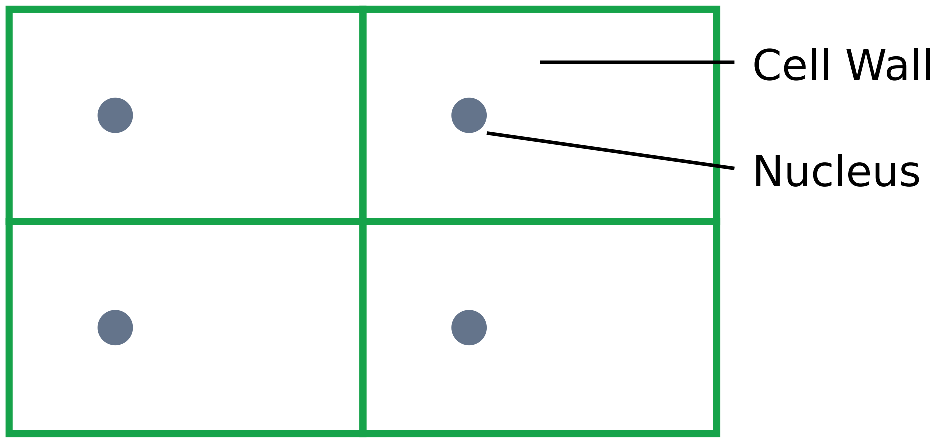
2. Dendrochronology (Tree Age Estimation)

Concept: In temperate and some tropical regions, one growth ring (annual ring) represents one year of growth.

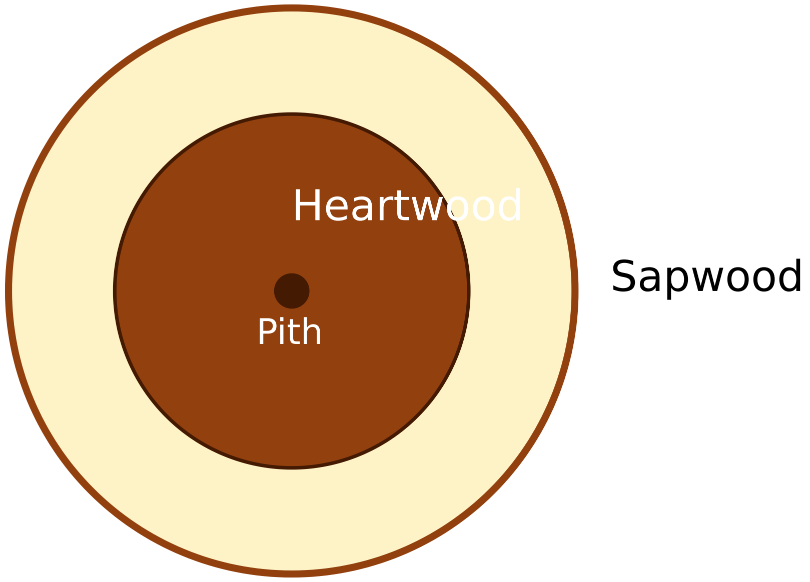
Calculation: Count the number of pairs of light wood (Earlywood) and dark wood (Latewood) from the pith (center) to the vascular cambium (bark side).

Formula: Age (years) ≈ Total Number of Annual Rings

Visual Library



Typical rectangular cells of Onion Peel as seen under a light microscope.



Cross-section of a woody stem showing Heartwood (dark, central) and Sapwood (light, outer).

1.1 Microscopic Observation

- **Algae (Spirogyra):** Filamentous green algae with spiral chloroplasts.
- **Fungi (Rhizopus):** Common bread mold; observe sporangia and hyphae.
- **Bryophyta (Moss):** Non-vascular plants; observe gametophyte and sporophyte.
- **Pteridophyta (Nephrolepis):** Ferns; observe sori on the underside of leaflets.

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1.2 Morphology of Root, Stem, and Leaves

- **Roots:** Tap root system (Dicot) vs Fibrous root system (Monocot).
- **Stems:** Modifications like Rhizome (Ginger), Tuber (Potato), Bulb (Onion).
- **Leaves:** Venation (Reticulate vs Parallel), Phyllotaxy (Arrangement of leaves).

2.1 Inflorescence & Flowers

Study of how flowers are arranged on the plant axis.

- **Racemose:** Main axis continues to grow (e.g., Mustard).
- **Cymose:** Main axis ends in a flower (e.g., Jasmine).
- **Flower Parts:** Calyx (Sepals), Corolla (Petals), Androecium (Stamens), Gynoecium (Carpels).

2.2 Morphology of Fruits

Type	Description	Example
Simple	From a single ovary	Mango, Tomato
Aggregate	From multiple ovaries of a single flower	Custard Apple
Multiple	From a whole inflorescence	Jackfruit, Pineapple

Study of vegetative and floral characters of key plant families relevant to wood and paper technology.

- **Annonaceae:** Example: Polyalthia (Ashoka tree).
- **Euphorbiaceae:** Example: Ricinus (Castor), Rubber tree.
- **Rutaceae:** Example: Citrus (Lemon).
- **Fabaceae:** Pea family; important for nitrogen fixation.
- **Myrtaceae:** Example: Eucalyptus (Crucial for paper industry).
- **Gramineae (Poaceae):** Grasses, Bamboo (Primary raw material for paper).

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4.1 Tissues and Cell Structure

Observation of Onion peel cells to understand basic plant cell structure (Cell wall, Cytoplasm, Nucleus).

4.2 Primary Structure

- **Dicot Stem:** Vascular bundles arranged in a ring; presence of cambium.
- **Monocot Stem:** Scattered vascular bundles; absence of cambium.

4.3 Secondary Thickening (Crucial for Wood Tech)

The increase in girth of the stem due to the activity of vascular cambium and cork cambium.

- **Heartwood (Duramen):** Central, dark-colored, non-functional, durable wood filled with resins/tannins.
- **Sapwood (Alburnum):** Outer, light-colored, functional wood that conducts water.
- **Tyloses:** Outgrowths from parenchyma cells into vessel lumens, often blocking them (common in heartwood).
- **Growth Rings:** Formed by the difference in Earlywood (Spring) and Latewood (Autumn) production.

Module-wise Practice Questions

Q1: Differentiate between Heartwood and Sapwood. (CO4)

Heartwood: Central part of the wood, dark in color, contains dead cells, resistant to decay, does not conduct water.
Sapwood: Peripheral part of the wood, lighter in color, contains living cells, susceptible to decay, active in water conduction.

Q2: What are Tyloses and why are they important in wood technology? (CO4)

Tyloses are balloon-like outgrowths of parenchyma cells into the lumen of xylem vessels. They block the vessels, making the wood less permeable to liquids. This is important in wood preservation and in making wood leak-proof for barrels.

Q3: Identify the family of Bamboo and its importance in the paper industry. (CO3)

Bamboo belongs to the family **Gramineae (Poaceae)**. It is a primary raw material for paper pulp due to its long fibers and rapid growth rate.

Q4: What is Dendrochronology? (CO4)

Dendrochronology is the scientific method of dating tree rings to determine the age of a tree and to study past environmental conditions.

Practice Model Practical Paper

Time: 3 Hours | Max Marks: 100

1. **Identification (20 Marks):** Identify the given specimens (A-E). Provide one identifying character for each. (Includes Algae, Fungi, Moss, Fern).
2. **Sectioning & Staining (30 Marks):** Take a T.S. of the given plant material (Dicot/Monocot Stem). Stain it and mount it. Draw a neat labeled diagram.
3. **Taxonomy (20 Marks):** Identify the family of the given flowering twig. List the vegetative and floral characters.
4. **Wood Anatomy (20 Marks):** Identify the wood sample provided. Comment on the presence of growth rings, heartwood/sapwood, and hardwood/softwood characteristics.
5. **Viva Voce & Record (10 Marks):** Based on the semester's practical work.

Quick Revision

- **Monocot Stem:** Scattered vascular bundles (like "monkey faces"). No secondary growth.
- **Dicot Stem:** Ring arrangement of vascular bundles. Shows secondary growth.
- **Annual Rings:** One ring = Early wood + Late wood. Used for age determination.
- **Hardwood:** Wood from Angiosperms (contains vessels).
- **Softwood:** Wood from Gymnosperms (lacks vessels, contains tracheids).
- **Phyllotaxy:** Arrangement of leaves on a stem (Alternate, Opposite, Whorled).

Glossary

Cambium
: A layer of meristematic tissue responsible for secondary growth (girth increase).

Inflorescence
: The arrangement of flowers on a floral axis.

Pith
: The central tissue of a stem, usually composed of parenchyma.

Venation
: The arrangement of veins in a leaf blade.

Xylem
: The vascular tissue responsible for water transport and providing mechanical support (Wood is secondary xylem).

Official Source Citation

This lesson is based on the official syllabus for **Course 2129: Botany Lab** under the **Revision 2021** curriculum provided by the State Institute of Technical Teachers' Training & Research (SITTTR), Kerala.

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